Juan C. Yamin

Department of Economics, Brown University
juan_yamin_silva@brown.edu | juancyamin.github.io | github.com/juancyamin

2026–27 Economics Job Market Candidate

Fields:	Econometrics, Development Economics
Research interests:	Statistical decision theory, empirical Bayes, experimental design, causal inference
Dissertation committee:	Toru Kitagawa, Soonwoo Kwon, Jonathan Roth

Education

Brown University, Providence, RI
Ph.D. in Economics — Expected May 2027
M.A. in Economics — May 2022

Universidad de los Andes, Bogotá, Colombia
M.Sc. in Economics — March 2019
B.Sc. in Economics — October 2017

Job Market Paper

Poverty Targeting with Imperfect Information

Targeted antipoverty programs in developing countries must rely on estimated rather than observed income, which leads to substantial targeting errors. I formulate the allocation of a fixed transfer budget based on those estimates as a statistical decision problem and show that the standard plug-in rule is inadmissible. I then develop a nonparametric empirical Bayes rule whose Bayes regret is bounded by the estimation error of the posterior mean poverty gaps and therefore inherits their rate of convergence. In simulations using household surveys from nine African countries, the rule reaches about 1.8 times as many poor people as the plug-in rule and matches its poverty-gap reduction with 6.7 percent less spending.

Other Working Papers

When and How to Pilot: Design Rules for Two-Wave Experiments

A pilot before the main wave yields variance estimates that can guide how to split the main sample between treatment and control. Balanced assignment ignores this evidence. Feasible Neyman allocation uses it, but with a small pilot it overreacts to noise and can do worse than balance. I propose a Conditional Minimax Regret (CMR) rule that minimizes worst-case regret over a finite-sample confidence set for the two variances. CMR retains balance's worst-case protection with high probability, converges to the Neyman allocation as the pilot grows, and attains the minimax-regret rate. In simulations calibrated to four field experiments, CMR avoids feasible Neyman's severe small-pilot losses while matching its large-pilot gains.

Two-Way Effects Models: A Nonparametric Empirical Bayes Approach
with Cole Davis — *Draft available upon request*

We develop a nonparametric empirical Bayes framework for models with unit and cluster effects. The framework allows unit effects to depend on latent cluster effects, relaxing independence restrictions in shrinkage estimation.

Publications

Birds of a Feather Collude Together: Subnational Alignment and Corruption

with Leopoldo Fergusson, Arturo Harker, and Carlos Molina
Conditionally accepted at the American Political Science Review

Using close elections in Colombia, we study how partisan alignment between municipal mayors and departmental governors affects corruption. Alignment increases the fabrication of "ghost" students, which inflates education transfers to municipalities, by 0.3 standard deviations, without improving genuine enrollment or student performance. It also raises discretionary hiring, patronage-based outsourcing, electoral fraud risk, and cross-level appointments of politicians' relatives. Aligned politicians nonetheless enjoy better future electoral prospects.

Software

cmrdesign (R and Python) — Documentation | Code | CRAN | PyPI

Uses pilot data to choose main-wave treatment and control sample shares and report a finite-sample regret certificate. Supports two-arm, multi-arm, and stratified experiments.

Research and Industry Experience

Brown University, Department of Economics June 2023–Jan. 2026
Research Assistant to Soonwoo Kwon, Providence, RI
Conducted simulation analyses and empirical applications for research on shrinkage estimation of fixed effects, inference in regression discontinuity designs under monotonicity, and bias-aware inference in regularized regression models.

Amazon, Delivery Experience May–Aug. 2025
Economist Intern, Bellevue, WA
Owned a research project evaluating a nationwide U.S. delivery-speed program. Formulated the causal question, acquired data using SQL, and implemented difference-in-differences and shift-share IV designs in Python to estimate direct and geographic spillover effects and quantify the program's total effect on customer delivery experience. Presented findings and recommendations to senior economists and program decision-makers.

Universidad de los Andes, Department of Economics Aug. 2017–July 2021
Research Assistant to Leopoldo Fergusson and James A. Robinson, Bogotá, Colombia
Built and linked administrative, archival, and web-scraped datasets for research on conflict, corruption, and state capacity. Implemented instrumental-variables and regression-discontinuity designs, examined mechanisms and heterogeneous effects, and conducted robustness and replication checks. Contributed empirical analysis and writing to Acemoglu, Fergusson, and Johnson's study of population growth and conflict, published in the *Review of Economic Studies*.

Selected Presentations

Midwest Econometrics Group Annual Meeting, Cincinnati 2026
Poverty Targeting with Imperfect Information (accepted for presentation)

World Congress of the Econometric Society, Seoul 2025
Poverty Targeting with Imperfect Information

Advances with Field Experiments, Chicago 2025
When and How to Pilot: Design Rules for Two-Wave Experiments

Awards and Fellowships

Merit Dissertation Fellowship, Department of Economics, Brown University 2026
International Travel Fund, Graduate School, Brown University 2025
Conference Travel Fund, Graduate School, Brown University 2025

Teaching Experience

Brown University, Teaching Assistant
Applied Econometrics II (Ph.D.), Peter Hull Spring 2026
Applied Econometrics I (Ph.D.), Toru Kitagawa Fall 2024
Using Big Data to Solve Economic and Social Problems (undergraduate), John N. Friedman Fall 2022 and 2023

Universidad de los Andes, Teaching Assistant Aug. 2018–June 2020
Advanced Econometrics (graduate), Raquel Bernal
Political Underpinnings of Prosperity and Poverty (graduate), James A. Robinson
Thinking Problems (undergraduate), Tomás Rodríguez

Academic Service

Organizer, Brown Metrics Coffee 2024–26

Technical Skills and Languages

Software: R, Python, Stata, MATLAB, SQL, Git, LaTeX
Languages: Spanish (native), English (fluent), French (fluent)

References

Toru Kitagawa
Professor of Economics
Brown University
toru_kitagawa@brown.edu

Soonwoo Kwon
Assistant Professor of Economics
Brown University
soonwoo_kwon@brown.edu

Jonathan Roth
Professor of Economics
Brown University
jonathan_roth@brown.edu